Signals Page — Complete Technical Specification

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This is the single consolidated reference document for the MindWealth Signals Page backend. It supersedes all prior individual notes on this topic. Read in full before implementation.

A. New fieldsDivyanshu/Ahil adds to C++ engine output

A1. Average loss return per trade

Add to Outstanding Signal and New Signals reports:

Column name: Backtested Returns (Lose Trades) [%] Max/Min/Avg

Symmetric with the existing Backtested Returns (Win Trades) column. Report maximum loss, minimum loss, and average loss across all backtested losing trades for each asset/function/interval/direction combination. Values are negative percentages. The Avg value is the critical one — it feeds directly into E[R].

A2. OscillatorDelta entry timing test – AHIL ONLY

We need to know whether there is a meaningful lag between the theoretical end of the divergence period and the observed signal date, and whether this lag affects performance.

Using whichever asset has the most OscillatorDelta trades in the backtest, run three scenarios:

- Scenario A: entry on exact observed signal date (current definition).
- Scenario B: entry 1 trading day earlier.
- Scenario C: entry 2–3 trading days earlier (approximate theoretical divergence end).

Record for each: win rate, avg win return, avg loss return, median return.

Also confirm: does the end of the thick black line in the OscillatorDelta visual represent the exact theoretical divergence end, or does the signal fire some days after that line ends?

Result determines how we set theoretical_entry_price for OscillatorDelta in the payload (see Section B3).

A3. IBKR B&H period — AHIL ONLY TO confirm

The report shows IBKR TrendPulse Daily with B&H CAGR = 27.95% over a stated 4-year window. Please confirm the B&H CAGR is computed over the same 4-year period as the strategy backtest, not a different window. If periods differ, the CAGR_diff is not a fair comparison.

B. New fields Divyanshu adds to payload and reports

B1. Asset class classification

Field name: asset_class

Derive from ticker using the map below. Needed to calibrate signal quality scoring correctly — bond ETFs and currencies have structurally lower returns than equities and must not compete on the same E[R] scale.

Exhaustive classification (Python): - PLEASE CHECK IF THIS IS EXHAUSTIVELIST OF ASSETS - 200

```
ASSET_CLASS_MAP = {
  # US equities
  'HDB':'equity','GS':'equity','BAC':'equity','META':'equity','MSFT':'equity',
  'IBKR':'equity','ORCL':'equity','CRM':'equity','HOOD':'equity','ZM':'equity',
  'ARM':'equity','ASML':'equity','SBUX':'equity','WMT':'equity','MA':'equity',
  'MCD':'equity','PG':'equity','GILD':'equity','BABA':'equity','TCEHY':'equity',
  'SFTBY':'equity','AMD':'equity','SPOT':'equity','WPM':'equity','BRK-B':'equity',
  'AMZN':'equity','INFY':'equity','NFLX':'equity',
  # Canada (.TO)
  'TFII.TO':'equity','EIF.TO':'equity','ATD.TO':'equity','AEM.TO':'equity',
  'KXS.TO':'equity','CSU.TO':'equity','NA.TO':'equity','BYD.TO':'equity',
  'SJ.TO':'equity','WFG.TO':'equity','WCN.TO':'equity','BNS.TO':'equity',
  'TD.TO':'equity','ATZ.TO':'equity','EFN.TO':'equity','WPM.TO':'equity',
  'PPL.TO':'equity','CNR.TO':'equity','DOL.TO':'equity','MX.TO':'equity',
  'XIU.TO':'equity_etf',  # Canadian index ETF
  # NZ (.NZ)
  'MFT.NZ':'equity','EBO.NZ':'equity','CNU.NZ':'equity','MEL.NZ':'equity',

'FRW.NZ':'equity','AIA.NZ':'equity','ATM.NZ':'equity','GNE.NZ':'equity','CEN.NZ':'equi
ty',
  # India (.NS)
  'HDFCBANK.NS':'equity','MAZDOCK.NS':'equity','SUNPHARMA.NS':'equity',
  'GRANULES.NS':'equity','NAUKRI.NS':'equity','BBTC.NS':'equity',
  'BEL.NS':'equity','BDL.NS':'equity','RELIANCE.NS':'equity',
  # Korea (treat as equity, not index)
  '005930.KS':'equity',
  # US equity ETFs
  'IWM':'equity_etf','SOXX':'equity_etf','VGT':'equity_etf','XLF':'equity_etf',
  'XLY':'equity_etf','XLU':'equity_etf','XLV':'equity_etf','JETS':'equity_etf',
  'ASHR':'equity_etf','FXI':'equity_etf','KWEB':'equity_etf','INDY':'equity_etf',
  'ENZL':'equity_etf','GDX':'equity_etf',
  # Commodity ETFs
  'GLD':'commodity_etf','SLV':'commodity_etf',
  # Bond ETFs
  'IEI':'bond_etf','TLT':'bond_etf','SHY':'bond_etf',
  # Crypto ETFs
  'IBIT':'crypto_etf',
  # Indices
  '^DJI':'index','^GSPC':'index','^NDX':'index','^NSEI':'index',
  '^TNX':'index','^BVSP':'index','^NZ50':'index',
  # Currencies
  'CAD=X':'currency','EURUSD=X':'currency',
  # Crypto
  'BTC-USD':'crypto','ETH-USD':'crypto',
}

def derive_asset_class(ticker):
    if ticker in ASSET_CLASS_MAP: return ASSET_CLASS_MAP[ticker]
    if ticker.endswith('.NS'): return 'equity'
```

```
if ticker.endswith(('.TO','.NZ','.KS')): return 'equity'
if ticker.startswith('^'): return 'index'
if '-USD' in ticker: return 'crypto'
if '=X' in ticker: return 'currency'
return 'equity'
```

B2. Computed quality fields

Compute and include these fields per signal in the payload. They feed the quality composite and the Claude JSON.

E[R] — the edge

The probability-weighted expected return per trade. This is the correct measure of signal edge, not win rate alone.

```
er = WR × avg_win_return + (1 − WR) × avg_loss_return
```

Until DIvyanshu adds avg_loss_return (Action A1), use the proxy:

```
avg_loss_proxy = -bt_max_loss(PLEASE CHECK THIS IS A POSITIVE NUMBER - HENCE ADDED THE NEGATIVE SIGN) × 0.6
er_proxy = WR × avg_win_return + (1 − WR) × avg_loss_proxy
```

Signal alpha per trade — does the signal beat random entry?

This separates two distinct causes of negative CAGR_diff (fully explained in Section D). It answers: does the signal identify entry windows that are better than a random entry on the same asset?

```
random_window_return = bt_bh(MEANS BUY AND HOLD)_cagr / 252 × bt_avg_hold_days
signal_alpha_per_trade = er − random_window_return
```

No new field from Divyanshu needed — both inputs already exist in the payload.

Interpretation:

- signal_alpha > +1%: timing skill is real — signal selects above-average windows
- signal_alpha near 0%: timing adds little above random entry
- signal_alpha < 0%: timing is poor — signal selects below-average windows

R:R ratio

```
rr_static  = avg_win_return / stop_distance_pct
             where stop_distance_pct = (current_price − nearest_stopMEANS STOP LOSS) / current_price × 100
rr_dynamic = (bt_avg_exit_price − current_price) / (current_price − nearest_stop) × 100
             (recomputed each day as price moves)
```

Include both. rr_static = original trade thesis. rr_dynamic = what remains from current price.

Theoretical entry price

Function	Theoretical entry price
TrendPulse	current_price / (1 + pct_above_breakout). Existing field '% Change vs Trendpulse Breakout day' gives pct_above_breakout.

FractalTrack	Track Level price (the Fibonacci retracement level) from existing 'Track Level/Price' column.
OscillatorDelta(AHIL CAN CHECK THIS FAST)	Signal price — pending Ahil's confirmation from Action A2. May update.
DeltaDrift / Sigmashell / BaselineDivergence / PulseGauge	Signal price.

Upside remaining % — X-axis for New Signals surface

```
bt_avg_exit_price = theoretical_entry_price × (1 + bt_avg_win_return / 100)
upside_remaining_pct = (bt_avg_exit_price − current_price)
                       / (bt_avg_exit_price − theoretical_entry_price) × 100
```

C. Signal quality composite — full formula

The quality composite is the Y-axis of the bubble chart. It has three components.

C1. Component 1 — E[R] (0 to 50 points)

```
er_score = clip(er / R_ref, 0, 1.0) × 50
```

R_ref is the reference E[R] for full marks. Set by asset class:

Asset class	R_ref	E[R] gate min	Example tickers
equity	10%	2.0%	HDB, TFII.TO, MSFT, META
equity_etf	4%	0.8%	IWM, XLF, SOXX, JETS
commodity_etf	6%	1.5%	GLD, SLV
bond_etf	1.5%	0.2%	IEI, TLT
crypto_etf	15%	3.0%	IBIT
index	3%	0.8%	^DJI, ^NSEI, ^NDX
currency	1.5%	0.3%	CAD=X, EURUSD=X
crypto	20%	4.0%	BTC-USD, ETH-USD

Example at R_ref = 10% for equity:

```
E[R] = 2%  → 2/10 × 50 = 10 pts
E[R] = 5%  → 5/10 × 50 = 25 pts
E[R] = 10% → capped at 50 pts (maximum)
E[R] = 21% → still capped at 50 pts
```

C2. Component 2 — Signal alpha (−15 to +15 points)

Replaces CAGR_diff in the composite. Measures timing skill above random entry.

```
alpha_score = clip(signal_alpha_per_trade / 5%, -1.0, +1.0) × 15
```

signal_alpha +5% → full +15 pts. signal_alpha −5% → full −15 pts. 0% → 0 pts.

Why 5%: a signal beating random entry by 5% per trade is exceptional timing. The clip at ±1.0 prevents outliers from dominating.

> *CAGR_diff is NOT in the composite. It is a diagnostic flag only — see Section D.*

C3. Component 3 — Sharpe (−6 to +8 points)

```
sharpe_score = clip((strategy_sharpe − 0.3) / 1.5, −0.3, +0.4) × 20
```

0.3 = minimum acceptable Sharpe. 1.8 = excellent (full marks). Asymmetric clips: max reward +8 pts, max penalty −6 pts.

At strategy Sharpe = 1.8: clip((1.8−0.3)/1.5, −0.3, +0.4) × 20 = 0.4 × 20 = 8 pts (maximum).

C4. Total composite

```
composite = er_score + alpha_score + sharpe_score
Range: approximately −21 (very poor) to +73 (exceptional)
NOT forced to 0–100. Signals ranked and colour-coded by composite value.
```

D. Why CAGR_diff is not a quality metric — and what it is used for

This section explains why CAGR_diff (strategy CAGR minus buy-and-hold CAGR) is not in the composite, and its two narrow remaining uses.

D1. The two separate facts inside CAGR_diff

A negative CAGR_diff can mean two completely different things:

Fact 1 — Opportunity cost of being out of the asset between signals

The strategy only deploys capital when the signal fires. Between signals, capital earns IRX. If the asset compounds steadily in the gaps, the annual strategy return suffers — not because the timing is bad, but because of the cash gaps. The signal is fine; the annual comparison is unfair.

Fact 2 — Poor timing: the signal selects below-average windows

The signal consistently enters during periods when the asset performs worse than its own historical average. A random entry would have done better.

Fact 1 response: hold the asset continuously — do not time it with this function.

Fact 2 response: avoid this signal/asset combination entirely.

CAGR_diff alone cannot tell you which one it is. Signal_alpha_per_trade resolves this.

> *In a live multi-signal portfolio, capital is never actually in cash between signals — it is redeployed into other signals. The true opportunity cost of not being in MSFT is the portfolio average return per unit of deployed capital, not MSFT buy-and-hold. CAGR_diff therefore overstates the opportunity cost for steadily compounding assets.*

> *CAGR_diff is also sensitive to the backtest start date. Starting at a cyclical peak depresses B&H CAGR and flatters CAGR_diff. Starting at a trough does the opposite. This is another reason not to use CAGR_diff as a quality measure — it varies with the accident of when the backtest window begins.*

D2. Worked examples

MSFT DeltaDrift Daily — Fact 1: opportunity cost, not bad timing

```
BT avg win return:     3.46% over avg 10 days
Win rate:              71.43%   Max loss: 8.2%
E[R] proxy:            0.7143 × 3.46 + 0.2857 × (−8.2 × 0.6) = +1.07%

B&H CAGR (4yr):        7.06%/yr
Random 10-day return: 7.06% / 252 × 10 = +0.28%
Signal alpha:          1.07% − 0.28% = +0.79%  ← positive: timing skill is real
CAGR_diff:             −2.72%                   ← looks bad but misleading
```

The signal picks MSFT entry windows that earn 0.79% more than a random 10-day entry. Timing is genuine. The negative CAGR_diff reflects only the compounding gap when capital is elsewhere between signals.

Display flag: 'Passive hold preferred — MSFT compounds steadily. Timing with DeltaDrift adds edge per trade but the cost of stepping in and out is high.'

JETS TrendPulse Daily — Fact 2: timing genuinely weak

```
BT avg win return:     2.84% over avg 17 days
Win rate:              81.82%   Max loss: 14.2%
E[R] proxy:            0.8182 × 2.84 + 0.1818 × (−14.2 × 0.6) = +0.77%

B&H CAGR (4yr):        8.63%/yr
Random 17-day return: 8.63% / 252 × 17 = +0.58%
Signal alpha:          0.77% − 0.58% = +0.19%  ← near zero: barely beats random
CAGR_diff:             −21.09%                  ← catastrophic
```

Signal alpha near zero means no meaningful timing skill. Catastrophic CAGR_diff combined with large max loss (14.2%) confirms this signal is destructive on JETS. Hard Gate A2d disqualifier.

Display flag: 'Avoid — signal barely beats random entry on JETS. Neither timing nor passive hold via this signal is supported.'

D3. What is a good E[R]?

E[R] cannot be judged in isolation. The benchmark is random_window_return for that asset and hold length. Signal_alpha_per_trade = E[R] − random_window_return is the answer.

Signal	E[R]	Random window	Signal alpha	Verdict
HDB DD Weekly	10.25%	0.56%	+9.69%	Exceptional
MSFT DD Daily	1.07%	0.28%	+0.79%	Decent — timing real, opportunity cost issue
JETS TP Daily	0.77%	0.58%	+0.19%	Thin — barely above random

IEI DD Weekly	−0.05%	0.02%	−0.07%	Negative — avoid

D4. The two remaining uses of CAGR_diff

- Hard Gate A2d: CAGR_diff below −15% (daily) / −20% (weekly) / −25% (monthly) AND signal_alpha near zero → disqualify from Tier A. Catches JETS-type situations.
- Passive hold flag: CAGR_diff negative AND signal_alpha positive → display note that the asset compounds well and may be better held continuously. Catches MSFT-type situations.

Outside these two uses, CAGR_diff does not influence signal ranking or composite scoring.

E. Gate logic

E1. Gate A2 — quality gates (all must pass for Tier A)

- A2a: E[R] ≥ asset-class minimum threshold (see R_ref table in C1).
- A2b: Forward Testing Win Rate ≥ 60%.
- A2c: Conviction ≥ minimum threshold (BQ + FS class assessment).
- A2d (nuanced): CAGR_diff ≥ −15% (daily) / −20% (weekly) / −25% (monthly). Waived if B&H CAGR is negative. For SHORT signals: benchmark is IRX cash rate, not B&H.
- A2e: R:R ≥ 1.0 — reward must exceed risk at BT avg exit.

E2. Gate A3 — lateness check (unchanged)

- Price progress ≥ 60% toward BT avg exit → LATE (unless dynamic R:R ≥ 3.0× waiver).
- Time elapsed > 150% of avg_holding_period → LATE (same waiver).
- Hard cutoffs from signal_date: Daily ≤ 5 trading days, Weekly ≤ 15, Monthly ≤ 25.

> *R:R in Gate A3 is dynamic R:R (remaining upside from current price), not static R:R at original entry. Include both rr_static and rr_dynamic as separate payload fields.*

F. What Claude receives — daily JSON payload

You are responsible for building this payload each morning. Claude does not compute quality scores — those are pre-computed. Claude's role is regime gate, conviction integration, cross-signal patterns, hard gate enforcement, and narrative generation.

F1. Per-signal fields

- symbol, function, interval, direction, signal_date, signal_price
- current_price, days_elapsed, mtm_pct
- bt_wr, fwd_wr, bt_avg_win_return, bt_avg_loss_return (or proxy), bt_max_loss
- bt_avg_hold_days, bt_strategy_sharpe, bt_strategy_cagr, bt_bh_cagr, cagr_diff
- nearest_target_price, nearest_stop_price

- theoretical_entry_price (per function formula in B2)
- upside_remaining_pct, er, signal_alpha_per_trade, composite_score
- rr_static, rr_dynamic
- asset_class
- conviction_bq_score, conviction_fs_class

F2. Regime context (per daily run)

- runic_combo_active: list of active Runic combos
- runic_on_watch: list of combos with one leg fired
- ssi_value: current SSI reading
- vix_level: current VIX

G. System constraints and other notes

G1. Portfolio universe constraint

When an asset is added to the portfolio (a position is opened), it must already exist in the approximately 200-asset model universe. If it does not, add it simultaneously. The portfolio is a strict subset of model-tracked assets.

G2. BT WR disclosure — add as footnote to reports

Add this footnote below the main table in both Outstanding and New Signals reports:

* BT WR is backtested across the full historical period for asset/function/interval/direction combinations meeting the 70% minimum threshold required for platform inclusion. This selection floor means BT WR cannot be compared to unconditional strategy win rates. Forward-test WR is the out-of-sample measure.

G3. Tests to run before release

- Verify E[R]_proxy is negative for IEI (bond ETF, avg_win 0.87%, max_loss 4.66%). Expected: −0.05%.
- Verify ^DJI upside_remaining_pct ≈ 40% (TrendPulse breakout field shows 2.57% already above breakout, consuming ~60% of the 1.66% BT avg return before signal was observed).
- Verify signal_alpha for JETS ≈ +0.19% and CAGR_diff = −21.09% triggers Gate A2d hard disqualifier.
- Verify rr_dynamic is recomputed from current_price daily when days_elapsed > 0.
- Verify asset_class for XIU.TO = equity_etf (not equity).

Questions: raise with Rohit.